

AI Medical Chatbot System Architecture

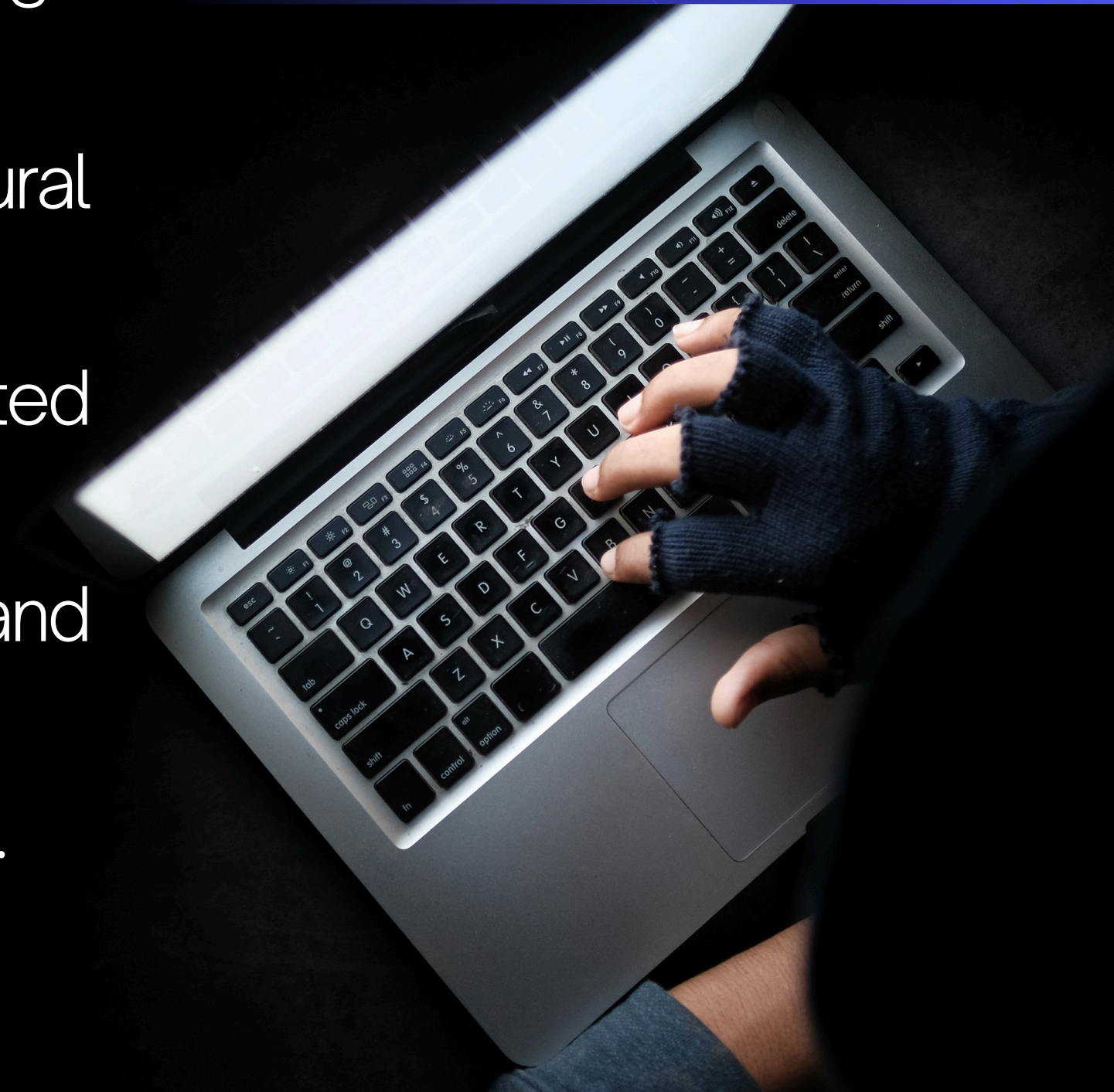
AI MEDICAL CHATBOT 2.0 – SYSTEM ARCHITECTURE

The AI Medical Chatbot 2.0 is a multi-modal conversational AI system that integrates speech recognition, image understanding, natural language reasoning, and speech synthesis to assist users in obtaining general health information.



KEY CHARACTERISTICS

- Multi-modal input: Accepts text, voice, and image queries.
- Multi-modal output: Provides answers in text and natural voice.
- Real-time processing: All responses are generated within seconds.
- Safety-first design: Detects emergency situations and advises immediate professional help.
- Privacy-aware: Does not store sensitive medical data.



Workflow

User Input

- Text Input: User types a question directly.
- Voice Input: User speaks into the microphone; system records audio.
- Image Input: User uploads a medical image (e.g., skin rash, X-ray).
- Multiple inputs can be given together (e.g., voice + image)

Workflow

Speech-to-Text

- If the user's input is voice, the system uses Whisper (via Groq API) to transcribe it into text.
- Whisper advantages:
 - High accuracy for medical terms.
 - Multi-language support for diverse users.
 - Noise-resistant in real-world conditions
- Output from STT is clean text ready for the LLM.

Workflow

Vision + Language Model
Processing

- **Core engine:** Uses a multimodal LLM like LLaMA-3 Vision.
- **Inputs to the LLM:**
 1. Text query (from typed input or STT transcription).
 2. Optional image (medical photo or report).

Workflow

Vision + Language
Model Processing50

How it works:

- Vision analysis: The model examines the uploaded image for visible patterns (e.g., skin lesions, abnormalities).
- Text reasoning: The model processes the query, understands context, and retrieves relevant knowledge.
- Combined reasoning: The LLM merges image findings with text context to produce a clear, concise answer.



Workflow



Safety Layer TTS Output

Workflow

SAFETY CHECK

The system scans the generated text for emergency keywords (e.g., “severe chest pain”, “difficulty breathing”).

If detected:

- Prepend a red-alert message advising urgent medical attention.
- Maintain calm, clear language to avoid panic.
- This layer ensures responsible AI behavior in healthcare contexts.

Workflow

Text-to-Speech (TTS)

- Converts the chatbot's final text answer into natural voice output.
- **Options:**
 1. gTTS: Free, good quality, supports many languages.
 2. ElevenLabs: Premium, human-like voice.
- Output is an MP3 or WAV file that plays in the browser.

Workflow

Output Stage & User Experience

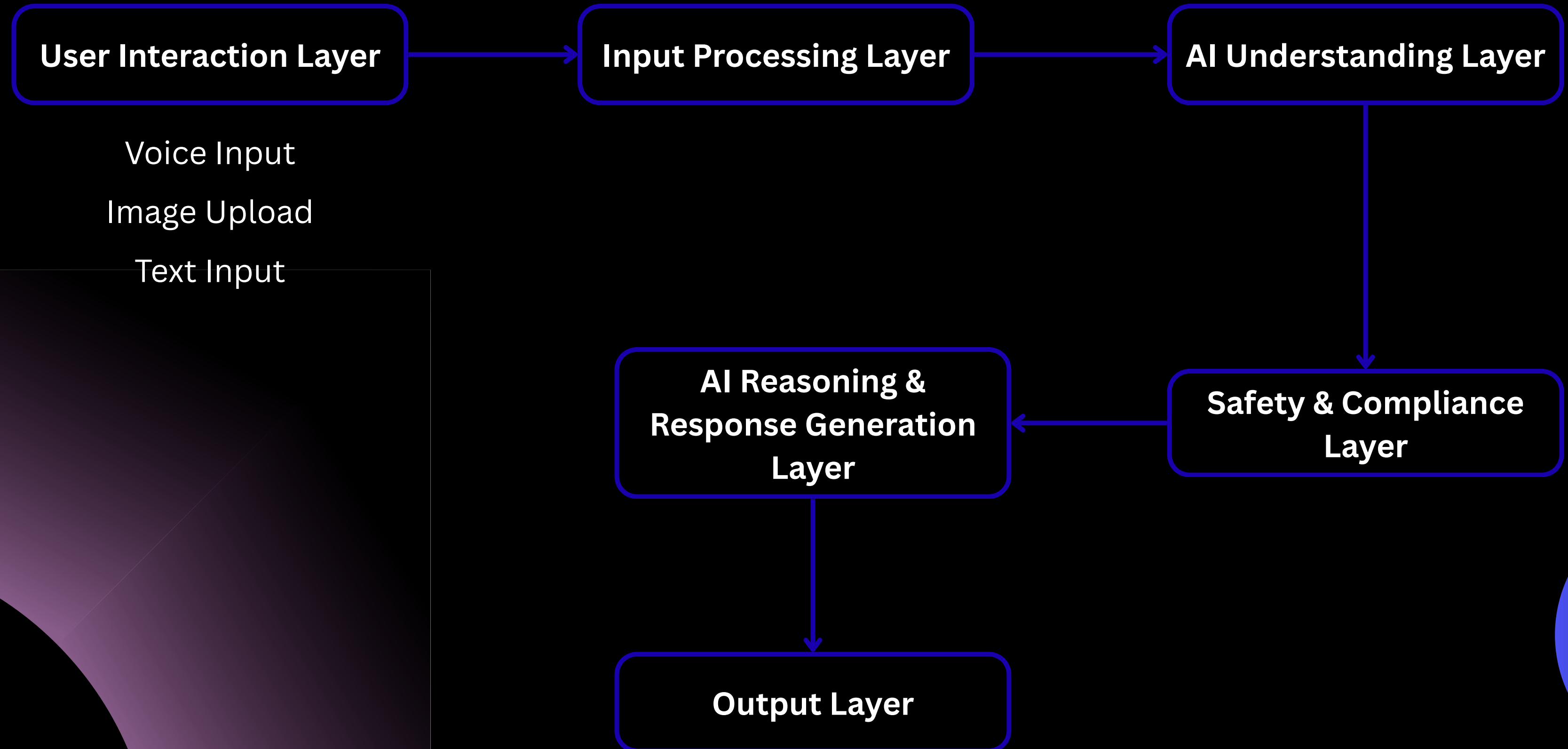
- **UI Display**

1. Text reply in a readable format.
2. Audio playback button for voice response.
3. Optional image preview if provided

- **Benefits of Gradio:**

1. Lightweight, browser-based.
2. Cross-platform (works on desktop & mobile).
3. No installation needed for users.

System Structure





Thank You

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